

Tasks on Jenkins-01

1. Install Jenkins and run Jenkins on port number 8081.

Implementation Steps

- Launch an Amazon Linux 2023 EC2 instance in AWS.
- Connect to the EC2 instance using SSH.
- Install Java 21: `sudo dnf install java-21-amazon-corretto -y`
- Add the Jenkins repository: `sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat/stable/jenkins.repo`
- Import the Jenkins repository key: `sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key`
- Install Jenkins: `sudo dnf install jenkins -y`
- Configure Jenkins to use port 8081: `sudo systemctl edit jenkins` Add: [Service] Environment="JENKINS_PORT=8081"
- Reload the systemd configuration: `sudo systemctl daemon-reload`
- Start Jenkins: `sudo systemctl start jenkins`
- Enable Jenkins to start automatically after reboot: `sudo systemctl enable jenkins`
- Check Jenkins service status: `sudo systemctl status jenkins`
- Allow TCP port 8081 in the EC2 Security Group.
- Access Jenkins using: <http://:8081>
- Complete the initial Jenkins setup and create the Jenkins administrator account.

instances (1) info						
less than a minute ago						
Saved filter sets						
Running						
Filter Instance by attribute or tag (case-sensitive)						
Instance state = running						
Clear filters						
☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	Jenkins-server	i-0abca851a71ffa378	Running	t3.micro	3/3 checks passed	us-east-1c

```
sec-user@ip-172-31-16-190 ~]$ sudo su -
[root@ip-172-31-16-190 ~]# sudo dnf update -y
amazon Linux 2023 Kernel Livepatch repository
dependencies resolved.
Nothing to do.
complete!
[root@ip-172-31-16-190 ~]# sudo dnf install java-17-amazon-corretto -y
last metadata expiration check: 0:00:18 ago on Tue Sep 29 12:28:07 2026.
dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
java-17-amazon-corretto	x86_64	1:17.0.20+10-1.amzn2023.1	amazonlinux	220
Installing dependencies:				
alsa-lib	x86_64	1.2.7.2-1.amzn2023.0.4	amazonlinux	503 k
cairo	x86_64	1.18.0-4.amzn2023.0.3	amazonlinux	717 k
dejavu-sans-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.3 M
dejavu-sans-mono-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	487 k
dejavu-serif-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.0 M
fontconfig	x86_64	2.13.94-2.amzn2023.0.2	amazonlinux	273 k
fonts-filesystem	noarch	1:2.0.5-12.amzn2023.0.2	amazonlinux	9.5 k
freetype	x86_64	2.13.2-5.amzn2023.0.2	amazonlinux	422 k
glib	x86_64	5.2.1-9.amzn2023.0.4	amazonlinux	49 k
google-noto-fonts-common	noarch	20240401-1.amzn2023.0.2	amazonlinux	17 k
google-noto-sans-vf-fonts	noarch	20240401-1.amzn2023.0.2	amazonlinux	593 k
graphite2	x86_64	1.3.14-7.amzn2023.0.3	amazonlinux	97 k
harfbuzz	x86_64	7.0.0-2.amzn2023.0.2	amazonlinux	873 k
java-17-amazon-corretto-headless	x86_64	1:17.0.20+10-1.amzn2023.1	amazonlinux	91 M
javapackages-filesystem	noarch	6.0.0-7.amzn2023.0.6	amazonlinux	12

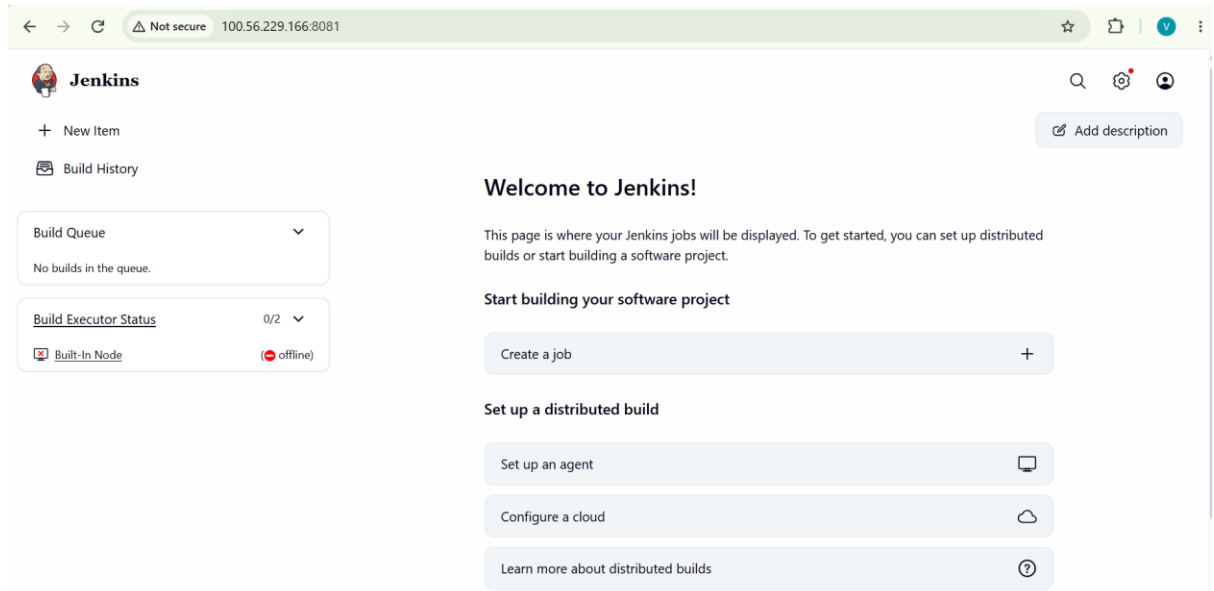
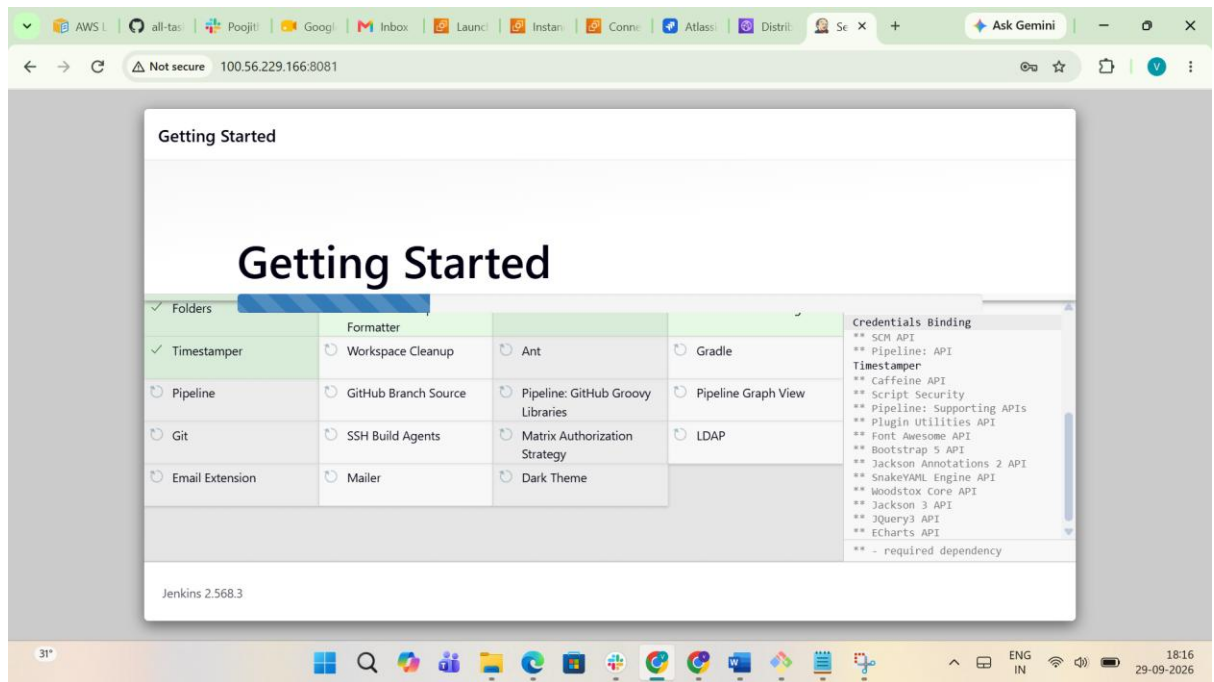
```
complete!
[root@ip-172-31-16-190 ~]# sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/redhat-stable/jenkins.repo
2026-09-12 12:28:53 --2026-09-12 12:28:53-- https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:79::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)[146.75.38.133]:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://pkg.jenkins.io/rpm-stable/jenkins.repo [following]
2026-09-12 12:28:53-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Reusing existing connection to pkg.jenkins.io:443.
HTTP request sent, awaiting response... 200 OK
length: 287 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'
```

```
[root@ip-172-31-16-190 ~]# sudo rpm --import https://pkg.jenkins.io/redhat/stable/jenkins.io-2023.key
[root@ip-172-31-16-190 ~]# sudo dnf install jenkins-y
Jenkins-stable
jenkins-stable
importing GPG key 0x14A8FC68:
  Userid : "Jenkins Project <jenkinsci-board@googlegroups.com>"
  Fingerprint: SE38 6EAD B55F 0150 4CAE 88CF 7198 F4B7 14AB FC68
  From : https://pkg.jenkins.io/rpm-stable/repoata/repodna/repodn.xml.key
jenkins-stable
Dependencies resolved.
```

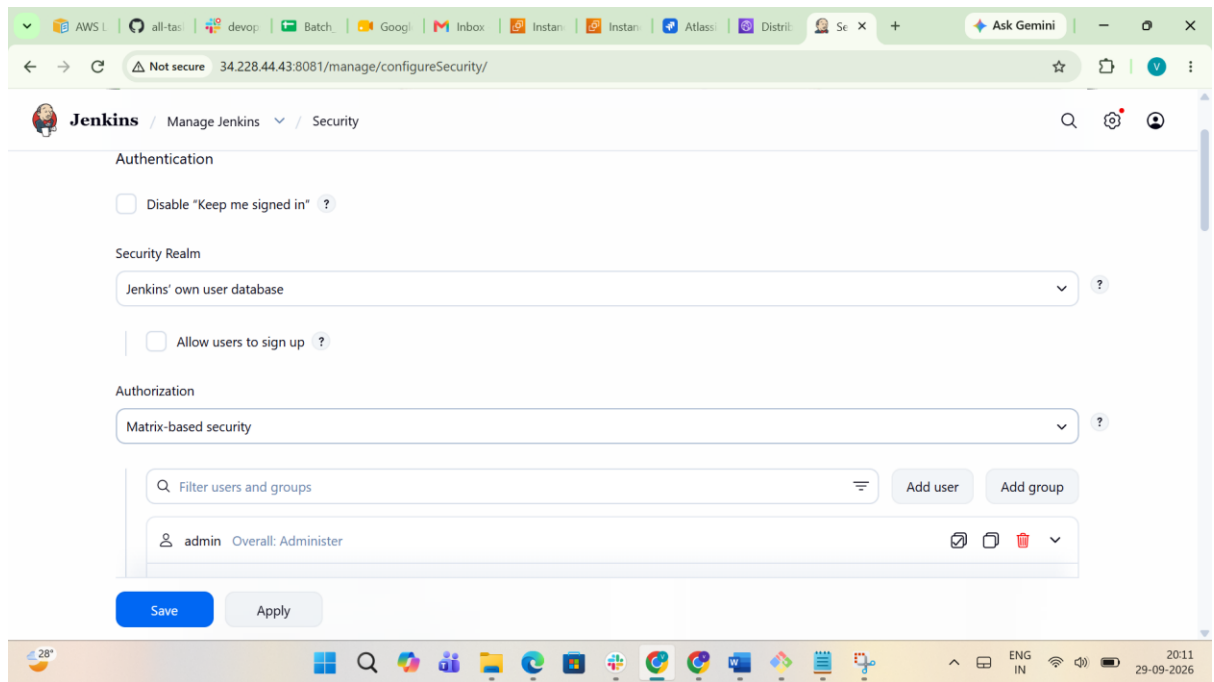
Package	Architecture	Version	Repository	Status
Installing: jenkins	noarch	2.568.3-1	jenkins	96
Transaction Summary				
Install 1 Package				

```
[root@ip-172-31-16-190 ~]# sudo cat /etc/systemd/system/jenkins.service.d/override.conf
[Service]
Environment="JENKINS_PORT=8081"
[root@ip-172-31-16-190 ~]# sudo systemctl daemon-reload
[root@ip-172-31-16-190 ~]# sudo systemctl start jenkins
[root@ip-172-31-16-190 ~]# cat /var/lib/jenkins/secrets/initialAdminPassword
bffd3363d2534593bc1f30230dc687f1
[root@ip-172-31-16-190 ~]# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/var/lib/systemd/system/jenkins.service; disabled; preset: disabled)
   Drop-In: /etc/systemd/system/jenkins.service.d
            └─override.conf
   Active: active (running) since Tue 2026-09-29 12:45:09 UTC; 1min 47s ago
   Main PID: 27345 (java)
     Tasks: 36 (limit: 1059)
    Memory: 583.2M
       CPU: 49.935s
    CGroup: /system.slice/jenkins.service
            └─27345 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8081

Sep 29 12:46:34 ip-172-31-16-190.ec2.internal jenkins[27345]: 2026-09-29 12:46:34.896+0000 [id=232] INFO jenkins.InitReactorRunner$1#onAttained: Listed
Sep 29 12:46:34 ip-172-31-16-190.ec2.internal jenkins[27345]: 2026-09-29 12:46:34.897+0000 [id=230] INFO jenkins.InitReactorRunner$1#onAttained: Prepared
Sep 29 12:46:34 ip-172-31-16-190.ec2.internal jenkins[27345]: 2026-09-29 12:46:34.897+0000 [id=231] INFO jenkins.InitReactorRunner$1#onAttained: Started
Sep 29 12:46:34 ip-172-31-16-190.ec2.internal jenkins[27345]: 2026-09-29 12:46:34.897+0000 [id=231] INFO jenkins.InitReactorRunner$1#onAttained: Augmented
Sep 29 12:46:34 ip-172-31-16-190.ec2.internal jenkins[27345]: 2026-09-29 12:46:34.898+0000 [id=230] INFO jenkins.InitReactorRunner$1#onAttained: System C
Sep 29 12:46:34 ip-172-31-16-190.ec2.internal jenkins[27345]: 2026-09-29 12:46:34.899+0000 [id=230] INFO jenkins.InitReactorRunner$1#onAttained: System C
Sep 29 12:46:34 ip-172-31-16-190.ec2.internal jenkins[27345]: 2026-09-29 12:46:34.899+0000 [id=230] INFO jenkins.InitReactorRunner$1#onAttained: Loaded a
Sep 29 12:46:34 ip-172-31-16-190.ec2.internal jenkins[27345]: 2026-09-29 12:46:34.900+0000 [id=230] INFO jenkins.InitReactorRunner$1#onAttained: Configur
Sep 29 12:46:34 ip-172-31-16-190.ec2.internal jenkins[27345]: 2026-09-29 12:46:34.900+0000 [id=230] INFO jenkins.InitReactorRunner$1#onAttained: Comple
Sep 29 12:46:34 ip-172-31-16-190.ec2.internal jenkins[27345]: 2026-09-29 12:46:34.900+0000 [id=791] INFO h.m.UpdateCenter$CompleteBatchJob$run: Completed
```



2. Secure Jenkins server



3. Create users called DevOps, Testing in Jenkins with Limited access.

Implementation Steps

- Open: Jenkins → Manage Jenkins → Users
- Click: Create User
- Create the first user: Username: DevOps
- Set the required password and user details.
- Create the second user: Username: Testing
- Set the required password and user details.
- Go to: Manage Jenkins → Security
- Open the Role-Based Authorization Strategy configuration.
- Configure roles and assign the required permissions to the users.
- Provide only the required Jenkins permissions rather than full administrative access.

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Jenkins

/ Jenkins' own user database

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Users 3

+ Create User

Username	Full name	
 admin	admin	
 DevOps	DevOps User	 
 Testing	Testing User	 

Jenkins 2.568.3

 DevOps User	Overall: Read · Agent: Build · Job: Build, Create · View: Read	   
 Testing User	Overall: Read · Agent: Create · Job: Build, Create	   

Dashboard - Jenkins

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← → ↻ Not secure 34.228.44.43:8081

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Jenkins

+ New Item

📁 Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

 Built-In Node

offline

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up your first job or build or start building a software project.

Start building your software project

Create a job

👤 DevOps User

⚙️ Theme

Light

📄 My Views

⚙️ Account

🖌️ Appearance

⚙️ Preferences

🔒 Security

🧪 Experiments

🔑 Credentials

REST API

Jenkins 2.568.3

34.228.44.43:8081/user/devops

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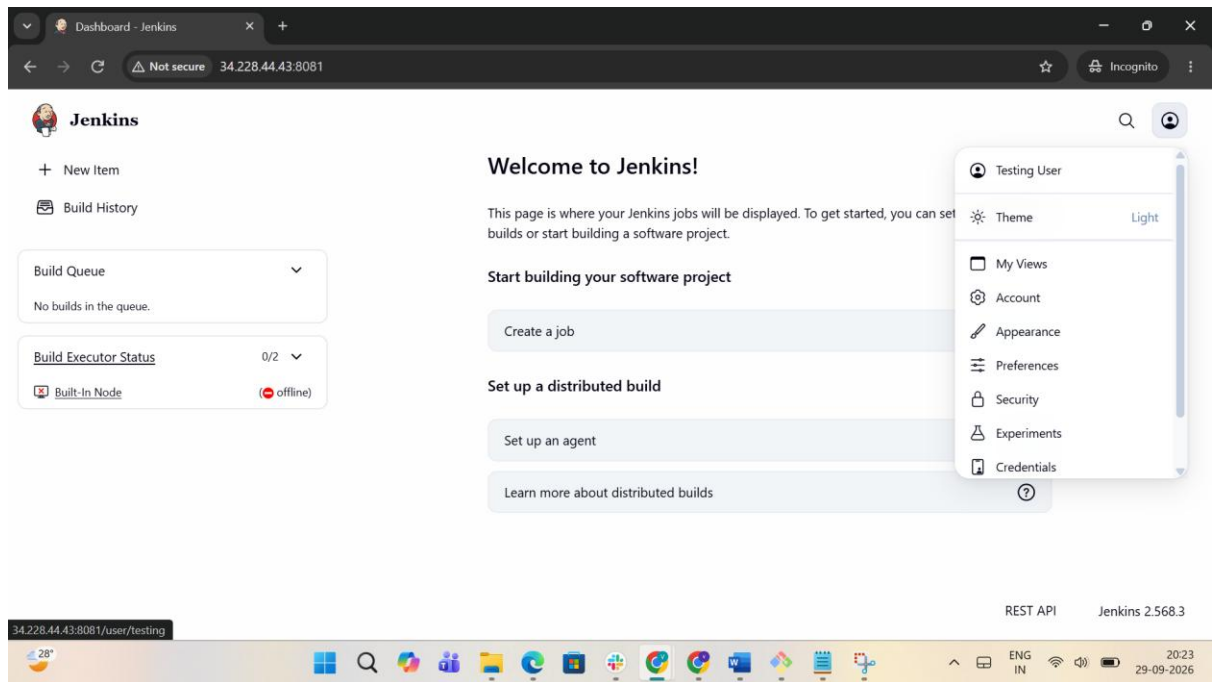
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20:22
29-09-2026



4. Configure labels and restrict the jobs to execute based on label only.

Implementation Steps

- Open: Jenkins → Manage Jenkins → Nodes
- Open the Built-In Node configuration.
- Add the following labels: master
- Save the node configuration.
- The same Built-In Node was used for the training environment, with both labels assigned to it.
- Create a Job Using the master Label
- Create a Freestyle project: first-Job
- Open the job configuration.
- Enable: Restrict where this project can be run
- Enter: master
- Add a simple build step if required.
- Save the job.
- Click Build Now.
- Verify that the job executes successfully.

Browser tabs: AWS L, all-tas, * dev, Batch, Googl, Inbox, Instan, Instan, Atlasi, Distri, Ask Gemini

Address bar: Not secure 34.228.44.43:8081/computer/(built-in)/configure

Jenkins / Nodes / Built-In Node / Configure

Status

Configure

Build History

Load Statistics

Script Console

Build Executor Status

Built-In Node

Number of executors ?
2

Labels ?
Master

Usage ?
Only build jobs with label expressions matching this node

Node Properties

- ☐ Disable deferred wipeout on this node ?
- ☐ Disk Space Monitoring Thresholds
- ☐ Environment variables

Save Apply

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Browser tabs: AWS L, all-tas, * dev, Batch, Googl, Inbox, Instan, Instan, Atlasi, Distri, fir, Ask Gemini

Address bar: Not secure 107.23.140.128:8081/job/first_job/configure

Jenkins / first_job / Configure

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

☐ Discard old builds ?

☐ GitHub project

☐ This project is parameterized ?

☐ Throttle builds ?

☐ Execute concurrent builds if necessary ?

☒ Restrict where this project can be run ?

Label Expression ?

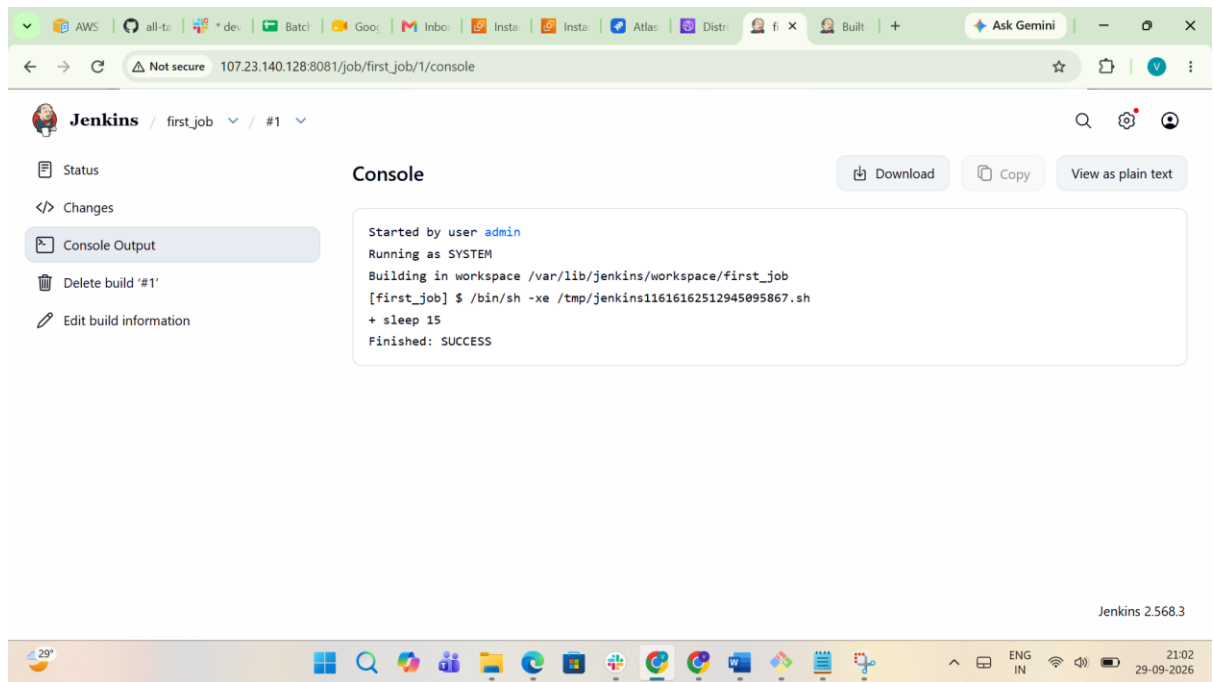
Master

Label Master matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced

Save Apply

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5. Create Three sample jobs using the below URL.

- **Source Code:**

https://github.com/betawins/Techie_horizon_Login_app.git

Prerequisite – Install Git in your server.

Job 1 — Techie-Horizon-Build

Implementation Steps

- Create a new Jenkins item: Techie-Horizon-Build
- Select: Freestyle project
- Enable: Restrict where this project can be run
- Under Source Code Management, select: Git
- Enter: https://github.com/betawins/Techie_horizon_Login_app.git
- Configure the branch: */master
- No credentials were required because the repository is public.
- Add the following build step:

```
echo "Starting build..."  
echo "Repository cloned successfully"  
ls -la  
echo "Build completed successfully"
```

- Save the configuration.
- Click Build Now.
- Verify that Jenkins successfully clones the repository.

Job 2 — Techie-Horizon-Test

- Create: Techie-Horizon-Test
- Select: Freestyle project
- Configure Git:
- Repository: https://github.com/betawins/Techie_horizon_Login_app.git
- Branch: */master
- Add the following build step:

```
echo "Starting application tests..."
```

```
ls -la
```

```
echo "Test execution completed successfully"
```

- Save the configuration.
- Click Build Now.
- Verify → Finished: SUCCESS

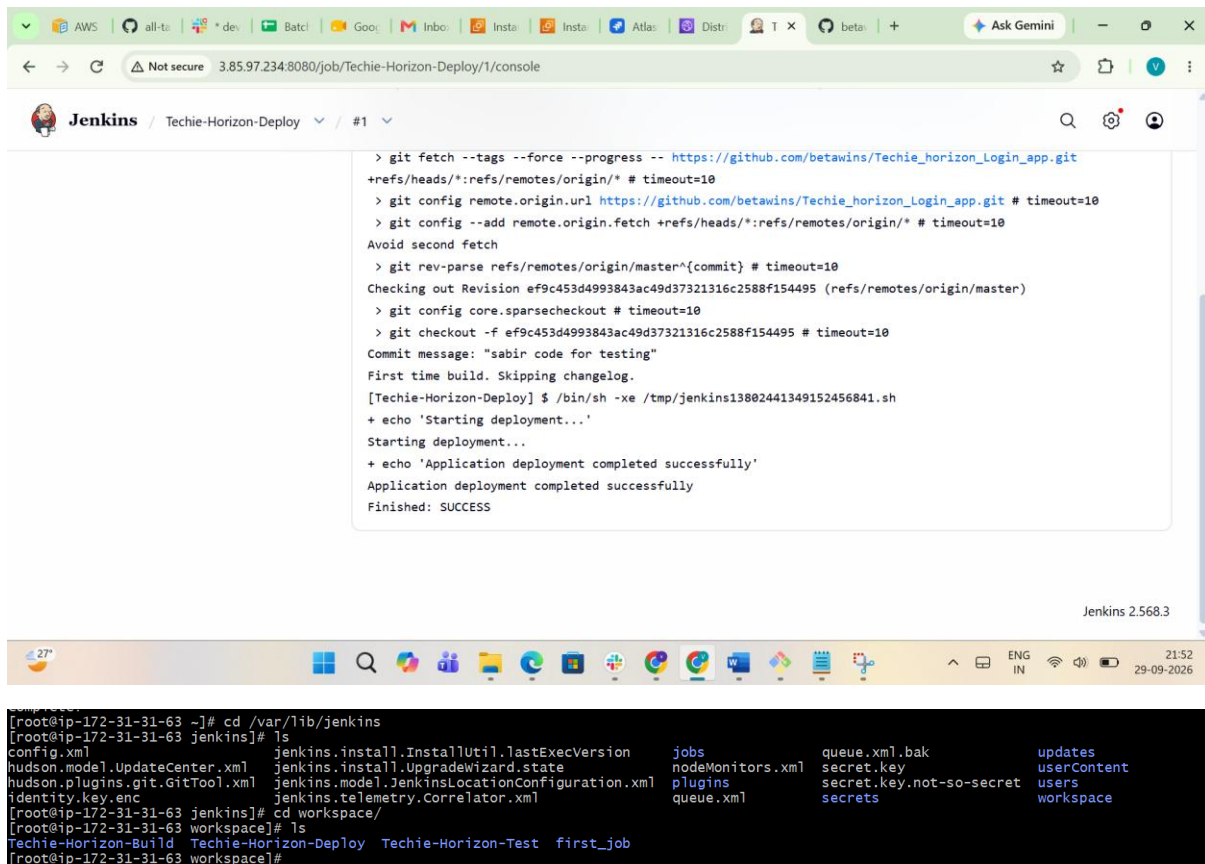
Job 3 — Techie-Horizon-Deploy

- Create: Techie-Horizon-Deploy
- Select: Freestyle project
- Configure the Git repository: https://github.com/betawins/Techie_horizon_Login_app.git
- Configure the branch: */master
- Add the following build step:

```
echo "Starting deployment..."
```

```
echo "Application deployment completed successfully"
```

- Save the configuration.
- Click Build Now.
- Verify the console output.



```
> git fetch --tags --force --progress -- https://github.com/betawins/Techie_horizon_Login_app.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/betawins/Techie_horizon_Login_app.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision ef9c453d4993843ac49d37321316c2588f154495 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f ef9c453d4993843ac49d37321316c2588f154495 # timeout=10
Commit message: "sabit code for testing"
First time build. Skipping changelog.
[Techie-Horizon-Deploy] $ /bin/sh -xe /tmp/jenkins13802441349152456841.sh
+ echo 'Starting deployment...'
Starting deployment...
+ echo 'Application deployment completed successfully'
Application deployment completed successfully
Finished: SUCCESS
```

```
[root@ip-172-31-31-63 ~]# cd /var/lib/jenkins
[root@ip-172-31-31-63 jenkins]# ls
config.xml          jenkins.install.InstallUtil.lastExecVersion  jobs          queue.xml.bak      updates
hudson.model.UpdateCenter.xml  jenkins.install.UpgradeWizard.state          nodeMonitors.xml  secret.key          userContent
hudson.plugins.git.GitTool.xml  jenkins.model.JenkinsLocationConfiguration.xml  plugins        secret.key.not-so-secret  users
identity.key.enc              jenkins.telemetry.Correlator.xml              queue.xml         secrets              workspace
```

```
[root@ip-172-31-31-63 jenkins]# cd workspace/
[root@ip-172-31-31-63 workspace]# ls
Techie-Horizon-Build  Techie-Horizon-Deploy  Techie-Horizon-Test  first_job
[root@ip-172-31-31-63 workspace]#
```

6. Create one Jenkins job using GitHub Private repository.

Step 1 — Create GitHub Personal Access Token

- Log in to GitHub.
- Navigate to: Settings → Developer settings → Personal access tokens → Tokens (classic)
- Generate a Personal Access Token.
- Configure the required repository access.
- For the classic token, enable: repo
- Generate the token.
- Copy and securely store the token.

Step 2 — Create Jenkins Credential

- Open: Jenkins → Manage Jenkins → Credentials
- Select: System → Global credentials
- Click: Add Credentials
- Select: Kind: Username with password
- Configure: Username:

- Password: GitHub Personal Access Token
- ID: github-private-repo
- Description: GitHub Private Repository
- Click Save.

Step 3 — Create the Private Repository Job

- From the Jenkins dashboard, click: New Item
- Enter: Techie-Horizon-Private-Repo
- Select: Freestyle project
- Click OK.

Step 4 — Configure Git

- Under Source Code Management, select: Git
- Repository URL: <https://github.com/vamshikumar-a/jenkins-private-app.git>
- Select the Jenkins credential: github-private-repo
- Branch: */main

Step 5 — Add Build Step

- Under: Build Steps → Execute shell
- The following commands were configured:

```
echo "Private repository cloned successfully"
echo "Files in workspace:"
ls -la
echo "Build completed successfully"
```

Step 6 — Build the Job

- Save the configuration.
- Open: Techie-Horizon-Private-Repo
- Click: Build Now
- Open the build number.
- Select: Console Output
- Verify that Jenkins successfully: Authenticated with GitHub. Cloned the private repository. Checked out the main branch. Executed the build commands. Displayed the

repository files. The successful build produced: • using credential github-private-repo and

- Private repository checkout successful
- Finally: Finished: SUCCESS

The image consists of two screenshots from a Windows desktop environment.

The top screenshot shows the GitHub 'Create new repository' page. The 'Owner' is 'vamshikumar-a' and the 'Repository name' is 'jenkins-private-app'. A message indicates 'jenkins-private-app is available.' The 'Description' field is empty. The 'Configuration' section is expanded, showing 'Choose visibility' set to 'Private', 'Add README' set to 'On', 'Add .gitignore' set to 'No .gitignore', and 'Add license' set to 'No license'.

The bottom screenshot shows the Jenkins 'Global' credentials page. The URL is '3.85.97.234:8080/manage/credentials/store/system/domain/_/'. The page title is 'Global' and it says 'Credentials that should be available everywhere.' There is a blue button '+ Add Credentials'. A single credential is listed: 'github-private-repo' with the username 'vamshikumar-a/*****' and the description 'GitHub Private Repository Credentials'.

At the bottom right of the Jenkins page, it says 'REST API' and 'Jenkins 2.568.3'.

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← → ↻ Not secure 3.85.97.234:8080/job/Techie-Horizon-Private-Repo/configure

Jenkins / Techie-Horizon-Private-Re... / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

None

Git ?

Repositories ?

Repository URL ?
https://github.com/vamshikumar-a/jenkins-private-app.git

Credentials ?
vamshikumar-a/***** (GitHub Private Repository Credentials) + Add

Advanced ▾

+ Add Repository

Save Apply

26° 22:04 29-09-2026

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← → ↻ Not secure 3.85.97.234:8080/job/Techie-Horizon-Private-Repo/2/console

Jenkins / Techie-Horizon-Private-Re... / #2 ▾

```
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision ce794e80c350840c2e1867c40e8398daec08ad27 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f ce794e80c350840c2e1867c40e8398daec08ad27 # timeout=10
Commit message: "Initial commit"
First time build. Skipping changelog.
[Techie-Horizon-Private-Repo] $ /bin/sh -xe /tmp/jenkins9934924347551912763.sh
+ echo 'Private repository cloned successfully'
Private repository cloned successfully
+ echo 'Files in workspace:'
Files in workspace:
+ ls -la
total 4
drwxr-xr-x. 3 jenkins jenkins 35 Sep 29 16:36 .
drwxr-xr-x. 8 jenkins jenkins 181 Sep 29 16:35 ..
drwxr-xr-x. 7 jenkins jenkins 146 Sep 29 16:36 .git
-rw-r--r--. 1 jenkins jenkins 21 Sep 29 16:36 README.md
+ echo 'Build completed successfully'
Build completed successfully
Finished: SUCCESS
```

26° 22:06 29-09-2026

✓ ☀ Techie-Horizon-Private-Repo ▾ 8.7 sec 2 min 34 sec 0.41 sec ▶

#3 #1